

Penrith Selective High School

2016

Higher School Certificate
Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A separate reference sheet is to be provided for this examination paper
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- All diagrams are not to scale
- Multiple choice answer sheet is on page 14 of this paper

Total Marks – 100

Section I Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–13

90 marks

- Attempt Questions 11–16
- Allow about 2 hour 45 minutes for this section

Student Number: _____

Students are advised that this is a trial examination only and cannot in any way guarantee the content or format of the 2016 Higher School Certificate Examination.

Section I:

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple choice answer sheet provided on page 14 for Questions 1–10.

Q1. Which expression is a correct factorisation of $x^3 - 8$

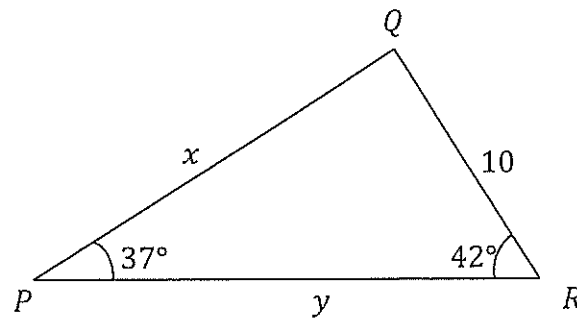
(A) $(x - 2)(x^2 - 2x + 4)$

(B) $(x - 2)(x^2 - 4x + 4)$

(C) $(x - 2)(x^2 + 2x + 4)$

(D) $(x - 2)(x^2 + 4x + 4)$

Q2. $\triangle PQR$ has side lengths x , y and 10 as shown. $\angle RPQ = 37^\circ$ and $\angle QRP = 42^\circ$.



Which of the following is correct for $\triangle PQR$?

(A) $x = 10 \times \frac{\sin 42^\circ}{\sin 37^\circ}$

(B) $y = 10 \times \frac{\sin 37^\circ}{\sin 101^\circ}$

(C) $x = \frac{10}{\sin 37^\circ}$

(D) $y = \frac{10}{\tan 37^\circ}$

- Q3. Which of the following values of m make the points $(4, -3)$, $(0, m)$ and $(-2, 5)$ collinear?
- (A) $m = 1$
 - (B) $m = \frac{7}{3}$
 - (C) $m = 4$
 - (D) $m = -\frac{1}{2}$
- Q4. The condition for the quadratic equation $3x^2 - 12x + k = 0$ to have real roots is
- (A) $k \leq 36$
 - (B) $k \geq 36$
 - (C) $k \leq 12$
 - (D) $k \geq 12$
- Q5. What is the centre and radius of the circle with the equation
- $$x^2 + y^2 + 6x - 8y - 11 = 0?$$
- (A) Centre $(-3, -4)$ and radius 36
 - (B) Centre $(-3, 4)$ and radius 36
 - (C) Centre $(-3, -4)$ and radius 6
 - (D) Centre $(-3, 4)$ and radius 6
- Q6. What is the equation of the normal to the curve $y = x^2 - 4x$ at $(1, -3)$?
- (A) $x + 2y - 7 = 0$
 - (B) $x - 2y - 7 = 0$
 - (C) $2x - y - 5 = 0$
 - (D) $2x + y + 5 = 0$

Q7. If $a > b$, which of the following is always true?

(A) $a^2 > b^2$

(B) $\frac{1}{a} > \frac{1}{b}$

(C) $-a > -b$

(D) $2^a > 2^b$

Q8. If $\tan 2x = \sqrt{3}$ in the domain $-180^\circ \leq x \leq 180^\circ$, the value of x is:

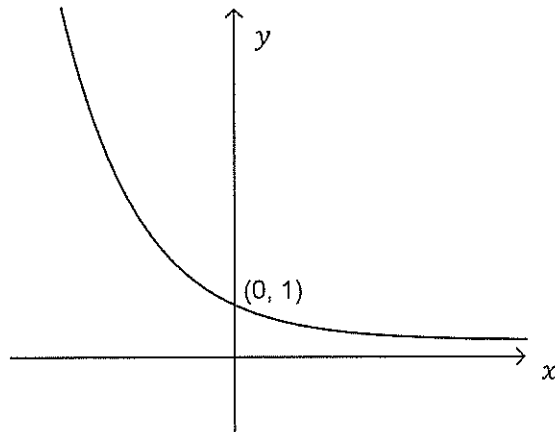
(A) $30^\circ, 210^\circ$

(B) $-150^\circ, -330^\circ$

(C) A and B

(D) None of above

Q9. The graph illustrated could be:



(A) $y = 2^x$

(B) $y = (-2)^x$

(C) $y = \left(\frac{1}{2}\right)^x$

(D) $y = \left(-\frac{1}{2}\right)^x$

- Q10. The sum of the interior angles of a regular polygon is 4140° . What is the size of each interior angle?
- (A) 157.5°
 - (B) 160°
 - (C) 162°
 - (D) 165.6°

Section II

90 Marks

Attempt Questions 11–16

Allow about 2 hour and 45 minutes for this section

Answer each question on a SEPARATE page.

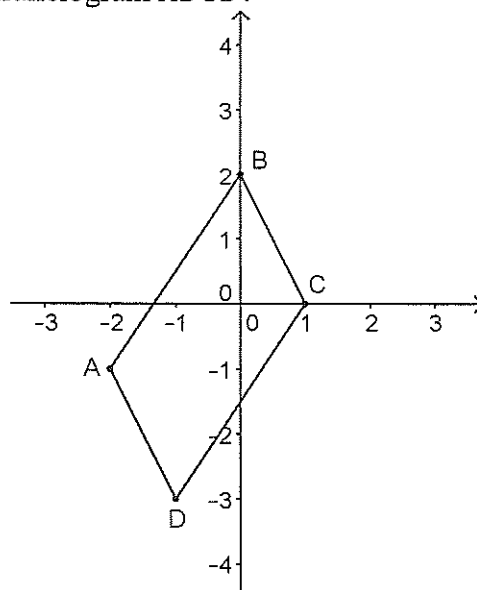
In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) **Start this question on a new writing page**

- a) Solve $|4x - 5| = 3(x + 1)$. 3
- b) Express $\frac{\sqrt{8}}{\sqrt{8} - \sqrt{7}}$ in the form $a + b\sqrt{14}$, where a and b are integers. 2
- c) Simplify $\frac{3^{m+1} - 3^m}{3^{2m+1} - 3^{2m}}$ 2

Question 11 continues on page 7

- d) In the diagram below, $A(-2, -1)$, $B(0, 2)$, $C(1, 0)$ and $D(-1, -3)$ are the vertices of a parallelogram $ABCD$.



- | | | |
|------|--|---|
| i) | Find the midpoint of CD . | 1 |
| ii) | Find the gradient of CD . | 1 |
| iii) | Show that the equation of CD is $3x - 2y - 3 = 0$. | 1 |
| iv) | Find the exact length of CD . | 2 |
| v) | Find the exact perpendicular distance from A to CD . | 2 |
| vi) | Hence find the area of parallelogram $ABCD$. | 1 |

End of Question 11

Question 12 (15 marks) **Start this question on a new writing page**

a) Show that $\cot \theta + \tan \theta = \operatorname{cosec} \theta \sec \theta$ 2

b) State the domain and range of $y = \sqrt{9 - x^2}$ 2

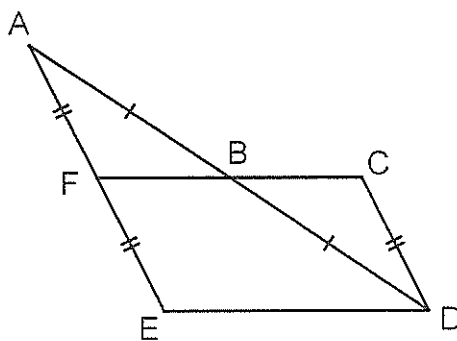
c) For the parabola: $(y - 2)^2 = 12(x + 3)$

i) State the coordinates of the vertex and the focus. 2

ii) Write down the equation of the axis of symmetry. 1

iii) Sketch this parabola showing all above information. 1

d) In the diagram, the line FC bisects AE at F and AD at B . The line AE is parallel to CD .



i) Prove that $\triangle ABF$ is similar to $\triangle ADE$. 2

ii) Hence explain why $ED = 2BF$. 1

Question 12 continues on page 9

e) If α and β are the roots of $5x^2 + 3x - 4 = 0$. Find the values of:

i) $\alpha + \beta$ 1

ii) $\alpha\beta$ 1

iii) $\frac{2}{\alpha} + \frac{2}{\beta}$ 1

iv) $\alpha^2 + \beta^2$ 1

End of Question 12

Question 13

(15 marks)

Start this question on a new writing page

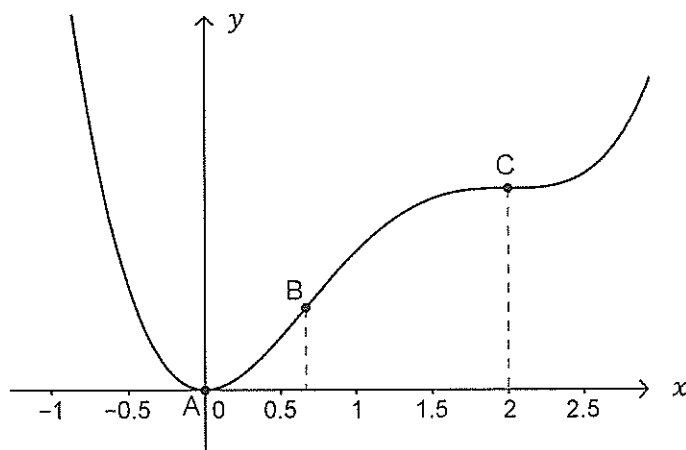
a) Given that $f(x) = (x^3 - 2)^5$, evaluate:

i) $f'(1)$ 2

ii) $f''(1)$ 2

b) The diagram shows the graph of a function $f(x)$. 3

The graph has a minimum turning point at A , a point of inflexion at B and a horizontal point of inflexion at C .



Sketch the graph of the derivative $f'(x)$.

c) Given that $y = x^2 - x$, show that $\frac{dy}{dx} - \frac{d^2y}{dx^2} = \frac{2y - x}{x}$ 3

d) Consider the curve given by $y = 3 + 21x - 9x^2 - x^3$

i) Find any turning points and determine their nature. 3

ii) Find any points of inflexion. 2

End of Question 13

Question 14 (15 marks) **Start this question on a new writing page**

- a) Find the 17th term of the sequence 5, 9, 13, 17, ... 1
- b) The sum of the first four terms of an arithmetic series equals to the 11th term. The sum of the 6th and the 7th term is -123 .
- i) Find the first term and the common difference. 2
- ii) Find the sum of the first 10 terms. 1
- c) i) In a geometric series $T_1 = 64$ and $T_4 = 1$. Find the common ratio. 1
- ii) Explain why this series has a limiting sum. 1
- iii) Find its limiting sum. 1
- d) Mario borrows \$650 000 to buy a house. An interest rate of 7.2% per annum, compounded monthly is charged on the outstanding balance. The loan is to be repaid in equal monthly instalments (M) over 30 year period. If A_n is the amount owing at the end of the n th month:
- i) Write down an expression for A_1 . 1
- ii) Show that the amount owing after three months is: 2
- $$A_3 = 650000(1.006)^3 - M(1 + 1.006 + 1.006^2)$$
- iii) Explain why $A_{360} = 0$. 1
- iv) Find the value of each instalment M to the nearest cent. 3
- v) Suppose now that Mario elects to pay \$5000 per month instead of the amount calculated in part iii). Show that he can pay off the loan 106 months earlier. 1

End of Question 14

Question 15 (15 marks) **Start this question on a new writing page**

a) Find

i) $\int \left(3x^2 - \frac{1}{x^2} \right) dx$ 2

ii) $\int (\sqrt{x} + 1)^2 dx$ 2

b) Find the area enclosed between the curve $y = x^2 - x - 6$, the x -axis and the lines $x = 1$ and $x = 4$. 2

c) Find the values of k if $\int_1^k (3x^2 - 25) dx = 24$ 3

d) Use Simpson's rule with 5 function values to find the approximate area enclosed between the curve $f(x) = \frac{x}{1+x^2}$, the x -axis and the line $x = 1$ and $x = 2$. Round your answer to 3 decimal places. 2

e) The region bounded by the curve $y = 2 - \sqrt{x}$ and the x -axis between $x = 0$ and $x = 4$ is rotated about the x -axis to form a solid.

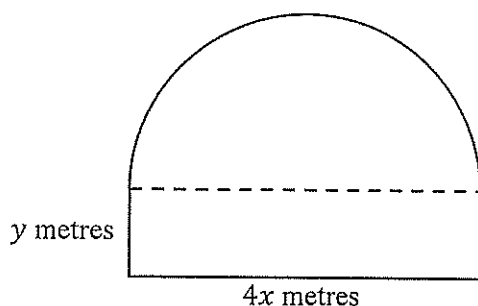
i) Sketch the above region, showing all important features. 1

ii) Hence or otherwise, find the exact volume formed. 3

End of Question 15

Question 16 (15 marks) **Start this question on a new writing page**

- a) Luigi has designed a garden bed which consists of a rectangle of width y metres and length $4x$ metres, and a semi-circle as shown in the diagram.



If the perimeter of the garden bed is to be 50 metres.

- i) Show that the perimeter of this garden bed can be expressed as 1

$$2x\pi + 2y + 4x = 50$$

- ii) Rearrange the above perimeter to express y in terms of x . 1

- iii) Show that the area of the garden bed can be given by the formula 2

$$A = 100x - 8x^2 - 2x^2\pi$$

- iv) Find the value of x that gives the maximum area. Correct your answer to 2 decimal places. 3

- b) Yoshi is on a paddle board in the ocean 3 km from the nearest point O on a straight beach. He needs to paddle to a point A east along the beach and walk further east the rest of the distance to meet his friend Peach who is 6 km along the beach from O. Yoshi is able to paddle at a rate of 4 km/h and walk at a rate of 5 km/h.

- i) Draw a diagram to represent this information. 1

- ii) Let x be the distance between point O and point A. Show that the total time $T(x)$ hours, for Yoshi to reach Peach is given by: 2

$$T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6 - x}{5}$$

- iii) Find value of x that gives the minimum time for Yoshi to reach Peach on the beach. 4

- iv) Find the minimum time for Yoshi to reach Peach on the beach. 1

End of Paper

Name: _____Antone_____

Teacher: _____Antone_____

Maths Trial 2016

4 Options Multiple Choice Answer Sheet for Qns 1 To 10

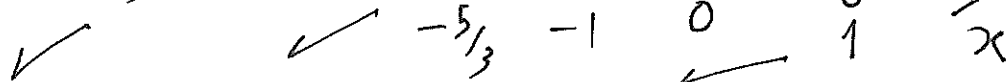
- | | | | | |
|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1) | ☒ A | B | C | D |
| 2) | A | B | C | ☒ D |
| 3) | A | ☒ B | C | D |
| 4) | A | B | C | ☒ D |
| 5) | A | B | ☒ C | D |
| 6) | A | ☒ B | C | D |
| 7) | A | ☒ B | C | D |
| 8) | A | B | C | ☒ D |
| 9) | A | ☒ B | C | D |
| 10) | ☒ A | B | C | ☒ D |

a) $2x^2 - 9x - 5 = (2x+1)(x-5)$ ✓✓

b) $|3x+1| < 4$

$$-4 < 3x+1 < 4 \Rightarrow -5 < 3x < 3$$

$$\Rightarrow -\frac{5}{3} < x < 1$$



c) $y = x^2 + 3x$, at $x=2$, $y=10$

$$y' = 2x + 3, \text{ at } x=2, m=7$$
 ✓

Equation of tangent at $(2, 10)$:

$$y - 10 = 7(x - 2)$$

$$\Rightarrow y = 7x - 4 \text{ or } 7x - y - 4 = 0$$
 ✓

d) $f(x) = x \sin 2x$

$$f'(x) = 2x \cos 2x + \sin 2x$$
 ✓

e) $x^2 = 8(y-3)$

$$a=2, \text{ Focus } (0, 5)$$
 ✓

f) $\theta^r = 20$ ✓

$$\ell = 2 \times 20 = 40 \text{ cm}$$
 ✓

g) $\int_1^4 e^{2x} dx = \frac{1}{4} [e^{2x}]_1^4$ ✓

$$= \frac{1}{4} (e^8 - e^2)$$
 ✓

D. Antonio

$$1) i) \frac{d}{dx} (e^{4x} + 2x)^3$$

$$= 3(e^{4x} + 2x)^2 (4e^{4x} + 2)$$

Some students did deduct 1 from the power

$$ii) \frac{d}{dx} \left(\frac{\cos x}{x+1} \right)$$

$$= \frac{-\sin x (x+1) - \cos x}{(x+1)^2}$$

Most students did this correctly

$$b) \int \frac{6x}{x^2+3} dx = 3 \int \frac{2x}{x^2+3} dx$$

$$= 3 \ln(x^2+3) + C$$

Most students did this correctly

$$c) i) T_{15} = a + (n-1)d$$

$$= 6 + (15-1)5$$

$$= 76$$

$$ii) S_{15} = \frac{15}{2} [2 \times 6 + (15-1)5]$$

$$= \frac{15}{2} (12 + 70)$$

$$= 615$$

Some students incorrectly stated the formula for T_n and not S_n .

$$iii) 350 = \frac{n}{2} (2 \times 6 + 700 = n(12 + 5n - 5))$$

$$700 = 5n^2 + 7n$$

$$5n^2 + 7n - 700 = 0$$

$$n = \frac{-7 \pm \sqrt{7^2 - 4 \times 5 \times -700}}{2 \times 5}$$

$$= \frac{-7 \pm \sqrt{49 + 14000}}{10}$$

$$n = -12.55 \text{ or } 11.53 \dots$$

$n > 0$ and must be an integer
 $\therefore n = 11$

$$d) i) A = \frac{h}{2} (0 + 2 \times (15 + 12 + 17) + 0) \text{ where } h = 5$$

$$= 220 \text{ m}^2$$

$$ii) V = 1.2 \times 220$$

$$= 264 \text{ m}^3$$

Some students did not solve for n .

Students did not get all y values. Some students did not know the trapezoid rule.

13 (i) A(0, 4) C(6, 1)

$$m = \frac{1-4}{6-0} = -\frac{3}{6} = -\frac{1}{2}$$

$$y - 4 = -\frac{1}{2}(x - 0)$$

$$2y - 8 = -x$$

$$x + 2y - 8 = 0$$

$$\begin{aligned} \text{(ii)} \quad d &= \frac{|1(1) + 2(-4) - 8|}{\sqrt{1^2 + 2^2}} \quad B(1, -4) \\ &= \frac{|1 - 8 - 8|}{\sqrt{5}} = \frac{15}{\sqrt{5}} \end{aligned}$$

most common error
mixing +ve and -ve

(iii) ~~m = 2~~, B(1, -4)

$$y + 4 = 2(x + 1)$$

$$y + 4 = 2x + 2$$

$$2x - y - 4 = 0$$

Solving simultaneously

$$2x - y - 4 = 0 \quad \dots (1)$$

$$x + 2y - 8 = 0 \quad \dots (2)$$

$$y = 2x - 4 \quad \dots (3)$$

Sub (3) into (2)

$$x + 2(2x - 4) - 8 = 0$$

$$x + 4x - 8 - 8 = 0$$

$$x + 4x = -10$$

$$5x = -10$$

$$x = -2$$

Sub into (1)

$$-4 - y - 4 = 0$$

$$-8 = y$$

$$\therefore (-2, -8)$$

many students did
not use simultaneous
equations but
tried to use
perpendicular
distance.

(13) b) (i) when $y=0$

$$6x - x^2 = 0$$

$$x(6-x) = 0$$

$$x=0 \text{ and } x=6$$

Students tended to integrate either

$$A = \int_0^7$$

$$\text{or } A = \int_0^6 + \int_6^7$$

$$(ii) A = \left| \int_6^7 (6x - x^2) dx \right|$$

$$= \left| \left[\frac{6x^2}{2} - \frac{x^3}{3} \right]_6^7 \right|$$

$$= \left| \left[3x^2 - \frac{x^3}{3} \right]_6^7 \right|$$

$$= \left| \left[3(7)^2 - \frac{(7)^3}{3} \right] - \left[3(6)^2 - \frac{(6)^3}{3} \right] \right|$$

$$= \left| \left[147 - 114\frac{1}{3} \right] - \left[108 - 72 \right] \right|$$

$$= 8\frac{2}{3} \text{ units}$$

$$c) (i) P(RR) = \frac{5}{7} \times \frac{4}{5} = \frac{20}{35} = \frac{4}{7}$$

$$(ii) P(\text{at least 1 W}) = 1 - P(RR) \\ = 1 - \frac{4}{7} \\ = \frac{3}{7}$$

$$(iii) P(\text{Both same}) = \left(\frac{5}{7} \times \frac{4}{5} \right) + \left(\frac{2}{7} \times \frac{1}{5} \right) \\ = \frac{22}{35}$$

14) (i) $y = x^3 + 3x^2 - 9x$
 $\frac{dy}{dx} = 3x^2 + 6x - 9$

when $dy/dx = 0$

$$3(x^2 + 2x - 3) = 0$$

$$3(x-1)(x+3) = 0$$

$$x = 1 \text{ and } x = -3.$$

$$\frac{d^2y}{dx^2} = 6x + 6$$

at $x = 1$, $d^2y/dx^2 = 12$

$$y = -5$$

$\therefore$ min turning point at $(1, -5)$

at $x = -3$, $y = 27$

and $d^2y/dx^2 = -12$

$\therefore$ max at $(-3, 27)$

(ii) $\frac{d^2y}{dx^2} = 6x + 6$

when $d^2y/dx^2 = 0$

$$6x + 6 = 0$$

$$6x = -6$$

$$x = -1$$

x	-2	-1	0
-----	------	------	-----

d^2y/dx^2	-6	0	6
-------------	------	-----	-----

(change in concavity)

$\therefore$ point of inflexion

at $(-1, 11)$

* Full coordinate was not given.

* Concave up/down was used in conclusion instead of max/min.

* many students did not test concavity

(iii) when $\frac{d^2y}{dx^2} < 0$

$$\therefore x < -1$$

(iv) $-6 \leq x \leq 4$

$$x = -6, y = -34$$

$$x = 4, y = 76$$

Some students
used max from
 $x = -3$ in part (i)

$\therefore$ max value at 76

$$\begin{aligned} \text{b) } V &= \int_a^b \pi y^2 dx \\ &= \pi \int_1^3 \frac{2x}{3x^2-1} dx \end{aligned}$$

$$= \frac{\pi}{3} \int_1^3 \frac{6x}{3x^2-1} dx$$

$$= \frac{\pi}{3} \left[\ln(3x^2-1) \right]_1^3$$

$$= \frac{\pi}{3} [\ln 26 - \ln 2]$$

$$= 2.69 \text{ units}^3$$

Some students
made careless
substitution errors.

c) (i) $M_0 = 100$

$$60 = 100e^{-35k}$$

$$\frac{60}{100} = e^{-35k}$$

$$\ln(0.6) = -35k$$

$$\therefore k = \frac{\ln(\frac{6}{10})}{-35}$$

Many students did not leave in exact form.

(ii) $M = M_0 e^{-kt}$

$$M' = -k M_0 e^{-kt}$$

where $M = M_0 e^{-kt}$

hence $M' = -kM$

$$M = 50g$$

$$M' = - \left(\frac{\ln \frac{6}{10}}{35} \right) \times 50$$

$$= -0.7297 \dots$$

$$\therefore M' = 0.73 \text{ g/min}$$

(iii) $S = 100e^{-kt}$

$$\frac{1}{20} = e^{-kt}$$

$$\ln(0.05) = -kt$$

$$t = \frac{\ln(0.05)}{-k} = 205.26$$

$$\therefore t = 206 \text{ min}$$

Suggested Solutions

Marker's Comments

a i. $\frac{T_3}{T_2} = \frac{T_2}{T_1} = \frac{2x}{3} \therefore \text{geometric}$

ii. $|r| < 1 \quad \left| \frac{2x}{3} \right| < 1 \quad -1 < \frac{2x}{3} < 1$

$$-3 < 2x < 3$$

$$-\frac{3}{2} < x < \frac{3}{2}$$

iii. $S_{\infty} = \frac{a}{1-r} = \frac{\frac{x}{3}}{1-\frac{2x}{3}} = \frac{\frac{x}{3}}{\frac{3-2x}{3}} = \frac{x}{3-2x}$

b i. $v(t) = \frac{6t(4+t^3) - 3t^2(3t^2)}{(4+t^3)^2}$

$$= \frac{24t - 3t^4}{(4+t^3)^2}$$

ii. rest when $v(t) = 0$

$$0 = \frac{24t - 3t^4}{(4+t^3)^2}$$

$$0 = 3t(8 - t^3)$$

$$3t = 0 \quad t^3 = 8$$

$$t = 0 \quad t = 2$$

(iii) $x(1) = \frac{3(1)^2}{4+(1)^3} = \frac{3}{5} \quad x(2+2\sqrt{2}) = \frac{3(2+2\sqrt{2})^2}{4+(2+2\sqrt{2})^2} = \frac{3}{5}$

$\therefore$ particle is in the same position at t_1 and t_2 .

(iv) the particle moves away from $x = \frac{3}{5}$ and then returns

(v) test

t	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{3}{2}$	2	$\frac{5}{2}$
$x'(t)$	0.8	0	0.69	0.38	0	-0.14
	$\nearrow$	$\leftrightarrow$	$\nwarrow$	$\nwarrow$	$\leftrightarrow$	$\searrow$

$\therefore$ max displacement when $x = 2$

$$x(2) = \frac{3(2)^2}{4+2^3} = 1$$

i. not enough to find $r = \frac{2x}{3}$ only

ii. some students found limiting sum instead of $|r| < 1$

iii. many students found sum of GP instead of limiting sum

b i. first line is enough for full marks without simplifying

(iii) 'show that' question $\rightarrow$ show substitution

(v) always test for max or min when it is mentioned

$$c i. A_1 = 20000 \times 1.0075 - P$$

$$A_2 = 20000 \times 1.0075^2 - 1.0075P - P$$

$$ii. A_n = 20000 \times 1.0075^n - P(1 + 1.0075 + \dots + 1.0075^{n-1})$$

$$A_n = 0$$

$$P = \frac{20000 \times 1.0075^n}{1 + 1.0075 + \dots + 1.0075^{n-1}}$$

$$= \frac{20000 \times 1.0075^n}{\frac{1.0075^{60} - 1}{0.0075}} = \$415.167$$

$$= \$415$$

ii. show the calculation
ie \$415.167

$$iii. 415 \times 60 = \$24910$$

$$24910 - 20000 = 4910 \text{ saved}$$

iii. show your calculation
ie subtracting
20000

$$a-i) \left(\frac{a+b}{5}\right)^2 = \frac{(a^2+b^2) + 2ab}{25} = \frac{23ab + 2ab}{25}$$

$$= \frac{25ab}{25} = ab \quad \checkmark$$

$$a-ii) \text{ Since } \left(\frac{a+b}{5}\right)^2 = ab$$

$$\Rightarrow \log \left(\frac{a+b}{5}\right)^2 = \log(ab)$$

$$\Rightarrow 2 \log \frac{a+b}{5} = \log a + \log b \quad \checkmark$$

$$\Rightarrow \log \frac{a+b}{5} = \frac{1}{2} (\log a + \log b)$$

many students
could not see
these simple
relations.

b-i)

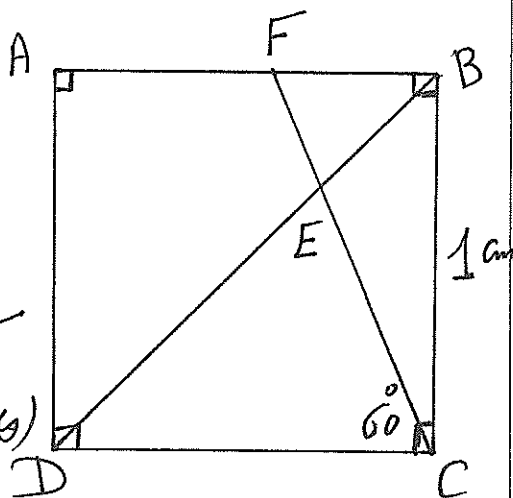
$$\angle CED = \angle BEF$$

(vertically opposite angles)

$$\angle DCE = \angle BFE \quad \checkmark$$

(Alternate angles on parallel lines)

$$\therefore \triangle BEF \parallel \triangle DEF \quad (\text{equi-angular})$$



$$b-ii) \frac{FB}{BC} = \tan 30 \Rightarrow FB = \frac{1}{\sqrt{3}} \quad \checkmark$$

$$b-iii) \triangle DEC : \triangle BEF$$

$$= (FB)^2 : (CD)^2 \quad (\text{Corresponding sides in similar } \Delta s)$$

$$= \frac{1}{3} : 1 = 1 : 3 \quad \checkmark$$

most students
could not see
this simple
relation.

$$\begin{aligned} \text{c-i)} \quad \Delta SOR &= \frac{1}{2} \times 4 \times 6 \sin \alpha \\ &= 12 \sin \alpha \quad \checkmark \checkmark \end{aligned}$$

$$\begin{aligned} \text{c-ii)} \quad \Delta S'OT &= \frac{1}{2} \times 2 \times 6 \sin(90 - \alpha) \\ &= 6 \cos \alpha \quad \checkmark \checkmark \end{aligned}$$

$$A = 12 \sin \alpha + 6 \cos \alpha$$

$$\text{c-iii)} \quad \frac{dA}{d\alpha} = 12 \cos \alpha - 6 \sin \alpha$$

$$\frac{dA}{d\alpha} = 0 \Rightarrow 12 \cos \alpha - 6 \sin \alpha = 0 \Rightarrow \tan \alpha = 2 \quad \checkmark$$

$$\frac{d^2A}{d\alpha^2} = -12 \sin \alpha - 6 \cos \alpha$$

When $\tan \alpha = 2$

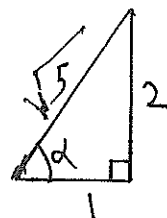
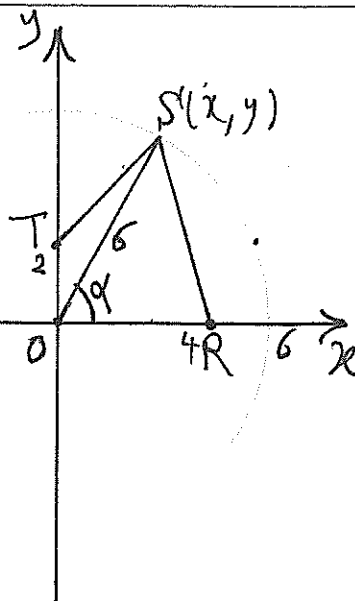
$$\frac{d^2A}{d\alpha^2} = -12 \times \frac{2}{\sqrt{5}} - 6 \times \frac{1}{\sqrt{5}} < 0$$

$\therefore$ There is a max value at $\alpha = \tan^{-1} 2$ $\checkmark$

c-iv) For $S(x, y)$:

$$x = 6 \cos \alpha = \frac{6}{\sqrt{5}} \quad \checkmark$$

$$y = 6 \sin \alpha = \frac{12}{\sqrt{5}} \quad \checkmark$$



D. Antone